

Program: Diploma in Fire Technology and Safety	
Course Code: 6462B	Course Title: Basic Fire Control Technology
Semester: 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To understand the fundamental principles of fire dynamics, combustion, and fire behavior.
 - To learn about the different types of combustion and their characteristics.
 - To familiarize students with fire growth stages, heat release rate, and heat transfer in fire scenarios.
- To study the operation, maintenance, and usage of fire service equipment and the principles behind fire hydraulics and fire ground operations.

Course Outcomes

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamental concepts of fire behavior, combustion types, and fire classification.	14	Understanding
CO2	Analyze the stages of fire growth and dynamics in enclosures, and understand heat release rates and transfer.	15	Analyzing
CO3	Operate and maintain essential fire service equipment and accessories used in firefighting.	13	Applying
CO4	Apply fire hydraulics concepts to calculate fire stream parameters, and understand fire ground operations.	16	Applying
	SeriesTest	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2				
CO2	3	3	3				
CO3	3	3	2	3			
CO4	3	3	3	3			

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the fundamental concepts of fire behavior, combustion types, and fire classification.		
M1.01	Introduction to fire: Temperature, heat, specific heat, flash point, fire point, ignition, and combustion.	3	Understanding
M1.02	Ignition sources: Pilot ignition, spontaneous ignition, and ignition sources.	4	Understanding
M1.03	Types of combustion: Rapid, spontaneous, physical explosion, chemical explosion, smoldering combustion.	4	Understanding
M1.04	Special kinds of combustion: Flash fire, pool fire, deep-seated fire, spillover, boil over, dust explosion, BLEVE, VCE.	3	Analyzing
CO2	Analyze the stages of fire growth and dynamics in enclosures, and understand heat release rates and transfer.		
M2.01	Introduction to enclosure fire dynamics: Description of fire growth stages in an enclosure (ignition, growth, flashover, fully developed fire, decay).	4	Understanding
M2.02	Heat release rate: Introduction, design fire, calorific value, fire load.	4	Understanding
M2.03	Factors controlling heat release rate in enclosure fires – burning rate, enclosure effects.	4	Analyzing

M2.04	Heat transfer in compartment fires: Measurement techniques, modes of heat transfer, and basic heat transfer equations.	3	Applying
	SeriesTestI	1	
Contents: Fire Growth Stages: Understanding the different stages of fire development in an enclosed space. Heat Release Rate: Study of factors like calorific value and fire load that control heat release rates. Heat Transfer: Overview of convection and radiation in fire and how heat transfer is measured. Fire Plumes and Flame Heights: Study of flame characteristics, including mean flame height, and the behavior of fire plumes.			
CO3	Operate and maintain essential fire service equipment and accessories used in firefighting.		
M3.01	Suction and delivery hoses, hose reel, hose fittings (couplings, adapters, branches, etc.).	4	Applying
M3.02	Introduction to fire fighting vehicles and appliances: Crash tenders, rescue tenders, hose laying tenders, control vans, hydraulic platforms.	3	Applying
M3.03	Ladders: Extension ladders, hook ladder, turntable ladders, snorkels.	4	Applying
M3.04	Miscellaneous fire service equipment: Fire pumps, primers, ropes, lines, and other small firefighting tools.	2	Applying
Contents: Fire Service Equipment: Study of the various firefighting tools, hoses, pumps, and their uses in emergency situations. Fire Fighting Vehicles: Introduction to different types of firefighting vehicles and their operation. Ladders and Platforms: Overview of the use and maintenance of ladders and hydraulic platforms used in fire service.			
CO4	Apply fire hydraulics concepts to calculate fire stream parameters, and understand fire ground operations.		
M4.01	Fire Hydraulics: Path and range of fire stream, nozzle types, and discharge capacity calculations.	5	Applying
M4.02	Pressure loss or gain due to elevation, back pressure, and flow in fire hoses.	4	Analyzing
M4.03	Water relay techniques and estimation of fire protection water requirements.	3	Applying

M4.04	Fire ground operations: Preplanning, action on arrival, control, and rescue methods.	4	Applying
	Series Test II	1	
Contents: Fire Hydraulics: Principles of fluid mechanics as applied to firefighting, including the calculation of nozzle discharge and pressure losses. Water Relay Techniques: Techniques used to transport water over long distances during firefighting. Fire Ground Operations: Overview of procedures followed during a fire emergency, including rescue, safety, and ventilation.			

Text/Reference:

T/R	Book Title/Author
R1	Hirst, Ron, Practical Fire Precautions, Gower Publishing Company Ltd., England, 1989.
R2	Jain, V.K., Fire Safety in Buildings (2nd Edition), New Age International (P) Ltd., New Delhi, 2010.
R3	Simon, Andrew L., Fire Hydraulics, Prentice-Hall Inc., USA, 1983.
R4	Karlsson, Bjorn & Quintiere, James G., Enclosure Fire Dynamics, CRC Press, USA, 2000.